Asia Quant Academy

Lecture 8 - Foundation Finance II

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Table of Content

Interest Rates

Fixed Income Markets

In fixed income markets, trading instruments are classified into two main types:

- Cash instruments, which include bonds,
- Derivative instruments like interest rate swaps (IRS), credit default swaps (CDS), and bond future, among others.

Bonds are debt instruments released by different entities, including governments, corporations, municipalities, and US Government Sponsored Enterprises, to raise capital in the financial markets.

Derivatives are synthetic instruments designed to capture particular characteristics of widely traded cash instruments, facilitating risk management and speculation.

Fixed Income Markets

- Every fixed income instrument represents a series of known or contingent cash flows. These cash flows are usually periodic and calculated based on a fixed or floating coupon applied to a principal amount (or notional principal).
- The market has established several conventions for applying the coupon to the appropriate accrual period. In the US dollar market, examples of these conventions include:
  1. the money market convention, which uses act/360 (actual days over a 360-day year),
  2. the bond market convention, which follows 30/360 (based on a 30-day month over a 360-day year).

LIBOR Rates

- The LIBOR (London Interbank Offered Rate), more accurately referred to as the ICE LIBOR, is the interest rate at which banks theoretically offer unsecured deposits to one another.
- Daily LIBOR fixings are released by Thompson Reuters on behalf of the Intercontinental Exchange (ICE) every London business day at 11:30 a.m. local time.
- These rates are determined based on quotes from a panel of participating banks. Detailed information regarding the composition of these panels and the calculation methods for the fixings can be found on the website www.theice.com/iba/libor.

LIBOR Rates

- LIBOR rates are not considered risk-free, although the banks involved typically have high credit ratings
- For each currency, LIBOR rates are quoted for seven different maturities: 1 day, 1 week, 1 month, 2 months, 3 months, 6 months, and 12 months.
- To simplify matters for this course, we will concentrate on the EUR market and refer to LIBOR specifically as the 3-month LIBOR rate.
- Besides spot transactions, various vanilla LIBOR-based instruments are actively traded on exchanges and over the counter (OTC), including:
  1. LIBOR futures,
  2. forward rate agreements,
  3. interest rate swaps

OIS Rates

- Following the 2008 credit crunch, LIBOR's reliability as a funding rate came under scrutiny.
- A significant part of the concern stems from the effective disappearance of the interbank unsecured lending market, which casts doubt on the accuracy of the quotes provided by participating banks to the BBA.
- Consequently, rates known as OIS rates, which are tied to the overnight rate set by the relevant central bank, gained prominence as benchmark funding rates. OIS stands for overnight indexed swap.

OIS Rates

- In the USD market, OIS rates are determined based on the daily effective federal funds rate. This rate holds significant economic importance.
- According to US law, banks must maintain specific minimum reserve levels, either as reserves held with the Fed or as vault cash. The required reserve level is influenced by the bank's outstanding assets and liabilities and is assessed at the end of each business day.
- If a bank's reserves fall below the required minimum, it must borrow funds to comply with regulations. The bank can obtain these funds from another bank with excess reserves or directly from the Fed at the "discount window."
- The interest rate paid by the borrowing bank to the lending bank is negotiated between them, and the weighted average of all such interest rates for transactions on a given day constitutes the effective federal funds rate for that day.

Forward Rate Agreement (FRA)

An FRA is basically a forward-starting loan, but without the exchange of the principal. The notional amount is simply used to calculate interest payments. By enabling market participants to trade today at an interest rate that will be effective at some point in the future, FRAs allow them to hedge their interest rate exposure on future engagements.

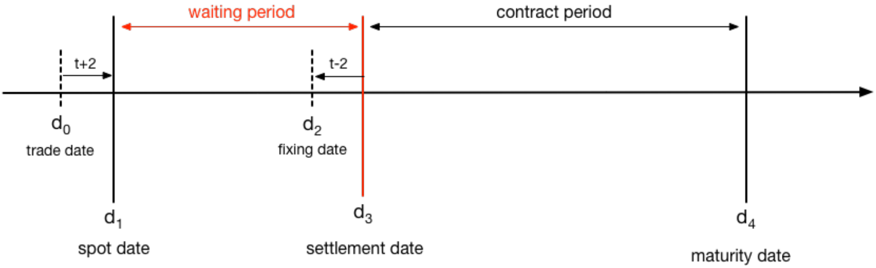


Figure: Cash flow of a typical FRA contract.

Forward Rate Agreement (FRA) - Settlement

Calculation of the interest differential
Interest differential $=$
$|$(Settlement rate $-$ Contract rate)$|$ $\times$ (Days in contract period/360) $\times$ Notional Amount

Calculation of the settlement amount
Settlement amount $=$
Interest differential/[1 $+$ Settlement rate $\times$ (Days in contract period/360)

The complete formula used to calculate the settlement amount is:

$$S = C \cdot \frac{(r_{set} - r_{rfa}) \cdot \dfrac{(d_{mty} - d_{set})}{d_{base}}}{1 + (\dfrac{d_{mty} - d_{set}}{d_{base}} \cdot r_{set})}$$

- If settlement rate >contract rate, FRA buyer receives the settlement amount
- If contract rate >settlement rate, FRA seller receives the settlement amount
- If settlement rate $=$ contract rate, no settlement amount is being paid

Forward Contract

Definition

A forward contract is an agreement to buy a specified quantity of a commodity or asset at a predetermined price on a future date. The price established today for this future transaction is known as the forward price. The purchaser of the underlying asset is referred to as being “long” on the forward contract.

Features

- Customized
- Non-standard and traded over the counter
- No money changes hands until maturity
- Non-trivial counterparty risk

Forward Contract - Example

Example

- Current price of soybeans is \$160/ton
- Tofu manufacturer needs 1,000 tons in 3 months
- Wants to make sure that 1,000 tons will be available
- 3-month forward contract for 1,000 tons of soybeans at \$165/ton
- Long side will buy 1,000 tons from short side at \$165/ton in 3 months

Future Contracts

Limitation of Forward

- Illiquidity
- Counterparty Risk

Definition

A futures contract is a standardized, exchange-traded agreement similar to a forward contract that is marked to market on a daily basis. This type of contract allows participants to take a long or short position in the underlying asset.

Future Contracts - Example

Yesterday, you bought 10 December live-cattle contracts on the CME, at a price of \$0.7455/lb.

- Contract size 40,000 lb
- Agreed to buy 40,000 pounds of live cattle in December
- Value of position yesterday:

$$(0.7455)(10)(40,000) = \$298,200$$

- No money changed hands
- Initial margin required (5%-20% of contract value)

Today, the futures price closes at \$0.7435/lb, 0.20 cents lower. The value of your position is

$$(0.7435)(10)(40,000) = \$297,400$$

which yields a loss of \$800.

Futures - SOFR

Definition
Three-Month SOFR (Secured Overnight Financing Rate) is a forward-looking interest rate that reflects the expected average SOFR over a three-month period.

Calculation
It is derived from market expectations implied by SOFR futures contracts and represents a cash-settled rate based on compounded daily SOFR during the contract reference quarter.

$$R = \prod_i (1 + \frac{d_i}{360} \times \frac{r_i}{100} - 1) \times \frac{360}{D} \times 100$$

Reference: CME SOFR-3M Documentation

Swap

The interest rate swap (IRS) market is by far the largest global derivatives market.

- A fixed for floating interest rate swap (or simply swap) is a transaction in which two counterparties agree to exchange periodic interest payments on a pre-specified notional amount.
- One counterparty (the fixed payer) agrees to pay periodically the other counterparty (the fixed receiver) a fixed coupon (say, 3.35exchange for receiving periodic LIBOR applied to the same notional.

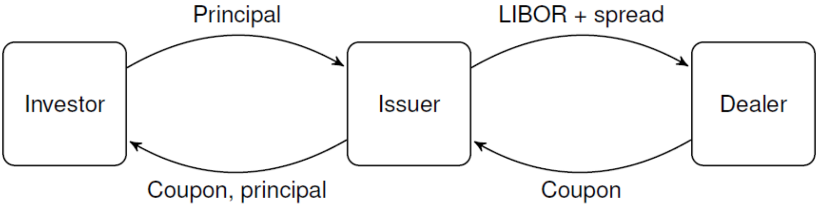


Figure: Cash flow of an LIBOR swap

Swap - Valuation

- Consider a simple swap with settlement at T_0 and matures at T. And, we assume the notional is \$1.
- Let $T_1^c < ... < T_{n_c}^c = T$ denote the coupon dates, let $0 \leq t \leq T_0$ denote the valuation date.
- The PV of the coupons on the fixed leg of the swap:

$$P_0^{fix}(t) = \sum_{j=1}^{n_c} \alpha_j C \cdot P_0(t, T_j^c)$$

where C is the coupon rate, $P_0(t, T_j^c)$ are the discount factors, and α_j are the day count fractions.

Swap - Valuation

- Let $T_1^f < ... < T_{n_f}^f = T$ denote the LIBOR payment dates.
- The valuation for the floating leg is:

$$P_0^{float}(t) = \sum_{1 \leq j \leq n_f} \delta_j L_j P_0(t, T_j^f)$$

- Where $L_j = L_0(T_{j-1}^f, T_j^f)$ is the LIBOR forward rate for settlement at T_{j-1}^f, $P_0(t, T_j^f)$ is the discount factor, and δ_j is the day count fraction.
- PV of a swap is

$$P_0(t) = P_0^{fix}(t) - P_0^{float}(t).$$

Swap - Valuation(LIBOR/OIS)

- Consider a LIBOR/OIS swap of maturity T, with payment dates on $T_1, ..., T_n = T$.
- The PV of the LIBOR leg is:

$$P_0^{LIBOR}(t) = \sum_{1 \leq j \leq n} \delta_j L_j P_0(t, T_j)$$

- while the PV of the OIS leg is:

$$P_0^{OIS}(t) = \sum_{1 \leq j \leq n} \delta_j (F_j + B) P_0(t, T_j)$$

  where B denotes the fixed basis spread for the maturity T.
- The break-even basis spread is thus given by

$$B_0(T_0, T) = \frac{\sum_j \delta_j (L_j - F_j) P_0(t, T_j)}{\sum_j \delta_j P_0(t, T_j)}$$

Options

Binomial Model

Let's start with equity option.
Here is the Cox-Ross-Rubinstein model,

$$\frac{S_{t+\Delta t} - S_t}{S_t} = \mu_t \Delta t + \sigma \sqrt{\Delta t}$$

where,

$$\epsilon_B = \begin{cases} 1, & \text{with probability } 1/2 \\ -1, & \text{with probability } 1/2 \end{cases}$$

is the Bernoulli random variable.

Binomial Model

Stock prices evolves as,

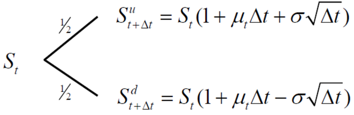


Figure: One-step price evolution of stock price

According to the model, the stock price should be,

$$
\begin{aligned}
S_{t,m} &= \frac{1}{1+r_t\Delta t}\left(\frac{1}{2}S^d_{t+\Delta t} + \frac{1}{2}S^u_{t+\Delta t}\right) \\
&= \frac{1}{1+r_t\Delta t}\left(\frac{1}{2}S_t(1+\mu_t\Delta t - \sigma\sqrt{\Delta t}) + \frac{1}{2}S_t(1+\mu_t\Delta t + \sigma\sqrt{\Delta t})\right) \\
&= \frac{1+\mu_t\Delta t}{1+r_t\Delta t}S_t \neq S_t \quad \text{unless } \mu_t = r_t
\end{aligned}
$$

Risk Premium

- The difference $\mu_t - r_t$ is called risk-premium in finance.
- It is the excess of return investors demand for making a risky investment.
- When building a stock price model, we must remove the risk premium. It is equivalent to say that we should take $\mu_t = r_t$.
- Then we have $S_{t,m} = S_t$.

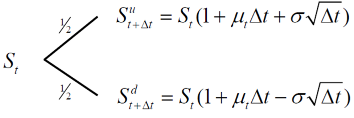


Figure: One-period maturity

Expectation pricing,

$$C_{0,0} = \frac{1}{1 + r_{0,0}\Delta t} \times (\frac{1}{2}C_{0,1} + \frac{1}{2}C_{1,1})$$

Is it correct?

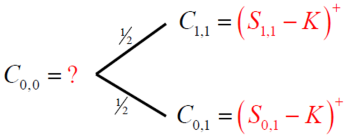


Figure: One-period option

Arbitrage Pricing

Consider replicating the payoffs with a portfolio of α and β units of shares and cash, s.t.

$$\alpha S_{0,1} + (1 + r_{0,0}\Delta t)\beta = C_{0,1}$$
$$\alpha S_{1,1} + (1 + r_{0,0}\Delta t)\beta = C_{1,1}$$

Solving the equation yields,

$$\alpha = \frac{C_{1,1} - C_{0,1}}{S_{1,1} - S_{0,1}}, \quad \beta = \frac{S_{1,1}C_{0,1} - S_{1,0}C_{1,1}}{(1 + r_{0,0}\Delta t)(S_{1,1} - S_{1,0})}$$

The value of the option is thus, $C_{0,0} = \alpha S_{0,0} + \beta$ which is unique.

Arbitrage Pricing - Linear Pricing Rule

Given the α, β, we may rewrite the option formula:

$$
\begin{aligned}
C_{0,0} &= \alpha S_{0,0} + \beta \\
&= \frac{C_{1,1} - C_{0,1}}{S_{1,1} - S_{0,1}} S_{0,0} + \frac{S_{1,1} C_{0,1} - S_{1,0} C_{1,1}}{(1 + r_{0,0}\Delta t)(S_{1,1} - S_{1,0})} \\
&= \text{...By rearranging and substitute the } S_{1,1} \text{ and } S_{0,1} \\
&= \frac{1}{1 + r_{0,0}\Delta t} \times \left(\frac{1}{2} C_{0,1} + \frac{1}{2} C_{1,1}\right)
\end{aligned}
$$

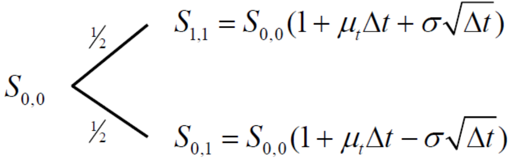


Figure: One-period maturity

Risk Neutral Pricing Measure

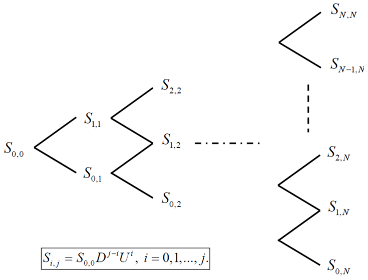
Rewrite the option formula into:

$$
\begin{aligned}
C_{0,0} &= \alpha S_{0,0} + \beta \\
&= \frac{C_{1,1} - C_{0,1}}{S_{1,1} - S_{0,1}} S_{0,0} + \frac{d(\Delta t)(S_{1,1} C_{0,1} - S_{1,0} C_{1,1})}{S_{1,1} - S_{0,1}} \\
&= d(\Delta t)\left(\frac{S_{1,1} - S_{0,0}/d(\Delta T)}{S_{1,1} - S_{0,1}} C_{0,1} + \frac{S_{0,0}/d(\Delta T) - S_{0,1}}{S_{1,1} - S_{0,1}} C_{1,1}\right) \\
&= d(\Delta T)(q C_{0,1} + (1-q) C_{1,1})
\end{aligned}
$$

where $0 < q < 1$.
The q, 1-q are called the risk-neutral pricing measure.

Assume the risk free rate is constant, then,

$$S^d_{t+\Delta t} = S_t[1 + r\Delta t - \sigma\sqrt{\Delta t}] = S_t D$$
$$S^u_{t+\Delta t} = S_t[1 + r\Delta t + \sigma\sqrt{\Delta t}] = S_t U$$



$$\boxed{S_{i,j} = S_{0,0} D^{j-i} U^i,\ i = 0,1,\ldots,j.}$$

Figure: Multi-period maturity

Multi-period Tree - Option

To price an option, one may use backward induction scheme.

For j = N-1:-1:0

For i=0:1:j

$$C_{i,j} = d(T_j, T_{j+1})(\frac{1}{2}C_{i,j+1} + \frac{1}{2}C_{i+1,j+1})$$

End

End

where $d(T_j, T_{j+1})$ discounts cash flow from T_{j+1} to T_j.

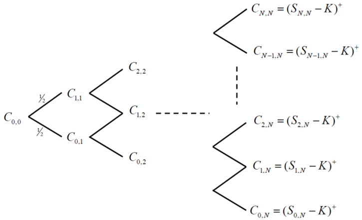


Figure: Multi-period maturity

Dynamical Hedging

In reality, options are hedged with dynamical hedging. At state (i,j), the seller fo the option need to keep:

$$\alpha_{i,j} = \frac{C_{1+1,j+1} - C_{i,j+1}}{S_{i+1,j+1} - S_{i,j+1}}$$

unit of shares for hedging.

Dynamical Hedging

By the backward induction,

$$C_{i,j+1} = \alpha_{i,j+1} S_{i,j+1} + \beta_{i,j+1}$$
$$C_{i+1,j+1} = \alpha_{i+1,j+1} S_{i+1,j+1} + \beta_{i+1,j+1}$$

Then, there is

$$\alpha_{i,j} S_{i,j+1} + (1 + r_j \Delta t)\beta_{i,j} = C_{i,j+1}$$
$$\alpha_{i,j} S_{i+1,j+1} + (1 + r_j \Delta t)\beta_{i,j} = C_{i+1,j+1}$$

It follows that,

$$\alpha_{i,j} S_{i,j+1} + (1 + r_j \Delta t)\beta_{i,j} = C_{i,j+1} = \alpha_{i,j+1} S_{i,j+1} + \beta_{i,j+1}$$
$$\alpha_{i,j} S_{i+1,j+1} + (1 + r_j \Delta t)\beta_{i,j} = C_{i+1,j+1} = \alpha_{i+1,j+1} S_{i+1,j+1} + \beta_{i+1,j+1}$$